OFFSIDE LABS

Dynamic Vault

Release 0.9.5

Smart Contract Security Assessment

October 2026

Prepared for:

Meteora

Prepared by:

Offside Labs

Sirius Xie



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1 About Offside Labs

Offside Labs is a leading security research team, composed of top talented hackers from both academia and industry.

We possess a wide range of expertise in modern software systems, including, but not limited to, *browsers*, *operating systems*, *IoT devices*, and *hypervisors*. We are also at the forefront of innovative areas like *cryptocurrencies* and *blockchain technologies*. Among our notable accomplishments are remote jailbreaks of devices such as the **iPhone** and **PlayStation 4**, and addressing critical vulnerabilities in the **Tron Network**.

Our team actively engages with and contributes to the security community. Having won and also co-organized *DEFCON CTF*, the most famous CTF competition in the Web2 era, we also triumphed in the **Paradigm CTF 2023** within the Web3 space. In addition, our efforts in responsibly disclosing numerous vulnerabilities to leading tech companies, such as *Apple*, *Google*, and *Microsoft*, have protected digital assets valued at over **$300 million**.

In the transition towards Web3, Offside Labs has achieved remarkable success. We have earned over **$9 million** in bug bounties, and **three** of our innovative techniques were recognized among the **top 10 blockchain hacking techniques of 2022** by the Web3 security community.

 **https://offside.io/**

 **https://github.com/offsidelabs**

 **https://twitter.com/offside_labs**

2 Executive Summary

Introduction

Offside Labs completed a security audit of *Meteora Dynamic Vault* smart contracts, starting on October 13th, 2026, and concluding on October 14th, 2026.

Project Overview

Meteora Dynamic Vault Release 0.9.5 adjusts deposits, withdrawals, and performance fee minting to use internal LP supply accounting for new and migrated vaults. Legacy vaults retain mint supply calculations, with permissionless migration initializing tracked supply from the current mint supply. Withdrawals require authorization from LP token account owners, and tracked supply is validated against mint supply. Version and supply fields reuse one strategy slot, reducing strategy capacity without increasing account size.

Audit Scope

The assessment scope contains mainly the smart contracts of the *vault* program for the *Meteora Dynamic Vault* project.

The audit is based on the following specific branches and commit hashes of the codebase repositories:

- Dynamic Vault
  - Codebase: `https://github.com/MeteoraAg/mercurial-vault`
  - Release 0.9.5: `PR-314`
  - Commit Hash: dcec1e6a83fa50c4c8c84254b1871706dc40493a

We listed the files we have audited below:

- Dynamic Vault Release 0.9.5
  - programs/vault/src/instructions/initialize_idle_vault.rs
  - programs/vault/src/instructions/migrate_vault.rs
  - programs/vault/src/instructions/mod.rs
  - programs/vault/src/instructions/remove_strategy.rs
  - programs/vault/src/lib.rs
  - programs/vault/src/state.rs

Findings

The security audit identified **no critical, high, medium, low, or informational issues** in the specified PRs. The implementation adheres to security best practices for the assessed components.

3 Detailed Summary

3.1 Scope Confirmation

The audit covered:

1. **Functionality**
   - LP supply accounting, deposit and withdrawal calculations, performance fee minting, withdrawal authorization, vault migration, and existing account layout compatibility..
2. **PRs**
   - PR Link: PR-314
     - Purpose: Introduce version based LP supply accounting and permissionless migration, adjust withdrawal ownership validation, and reuse existing account space for version and supply fields.

3.2 Conclusion

After thorough review, the audited components:

- **Follow secure design patterns** for the intended functionality covered by the audit;
- **Introduce no new security risks** within the given threat model.

4 Additional Notes

4.1 Assumptions

Integrators will support version based vault LP supply accounting before vault migration.

4.2 Review Logs

- Confirmed that the 32 bytes reused from `Vault.strategies[29]` were zero across all inspected vault accounts.
- Confirmed that no Dynamic Vault LP tokens were held in token accounts owned by the System Program or the incinerator at the inspected snapshot.
- The team confirmed that it will proactively migrate existing vaults using the planned migration script.

5 Disclaimer

This audit report is provided for informational purposes only and is not intended to be used as investment advice. While we strive to thoroughly review and analyze the smart contracts in question, we must clarify that our services do not encompass an exhaustive security examination. Our audit aims to identify potential security vulnerabilities to the best of our ability, but it does not serve as a guarantee that the smart contracts are completely free from security risks.

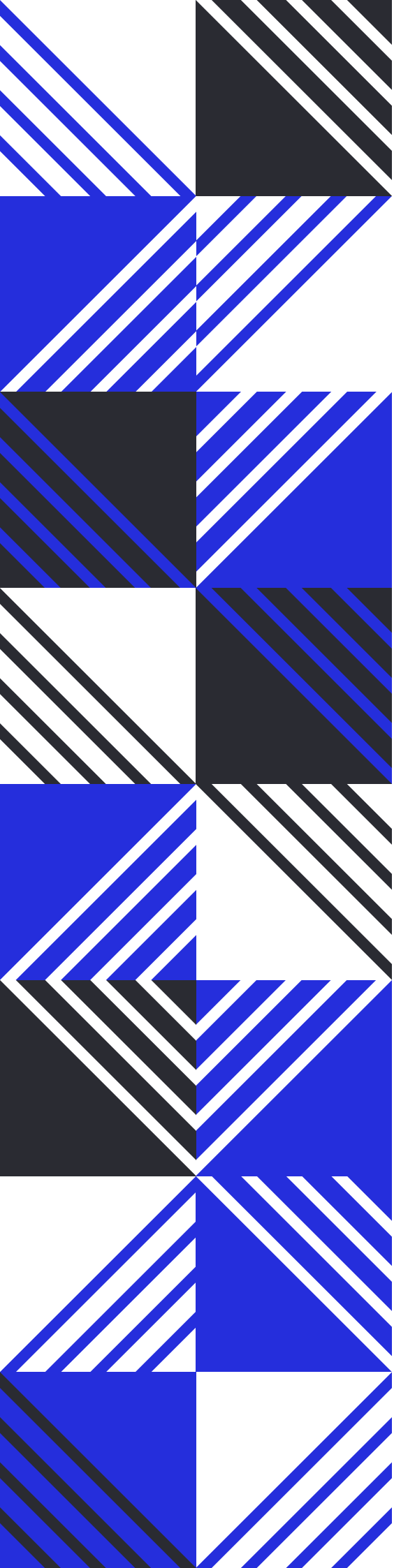
We expressly disclaim any liability for any losses or damages arising from the use of this report or from any security breaches that may occur in the future. We also recommend that our clients engage in multiple independent audits and establish a public bug bounty program as additional measures to bolster the security of their smart contracts.

It is important to note that the scope of our audit is limited to the areas outlined within our engagement and does not include every possible risk or vulnerability. Continuous security practices, including regular audits and monitoring, are essential for maintaining the security of smart contracts over time.

Please note: we are not liable for any security issues stemming from developer errors or misconfigurations at the time of contract deployment; we do not assume responsibility for any centralized governance risks within the project; we are not accountable for any impact on the project's security or availability due to significant damage to the underlying blockchain infrastructure.

By using this report, the client acknowledges the inherent limitations of the audit process and agrees that our firm shall not be held liable for any incidents that may occur subsequent to our engagement.

This report is considered null and void if the report (or any portion thereof) is altered in any manner.



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